

Review 3 (Final Report)
Artificial Intelligence(CSE3013)



**HEART DISEASE PREDICTION USING VARIOUS
CLASSIFICATION ALGORITHMS AND THEIR
COLLECTIVE ANALYSIS**

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i) Problem Statement:

To predict if a person has heart disease using different types of classification algorithm. We are planning to use logistic regression, K means, Naive Bayes, SVM and KNN algorithm. Classification algorithms are used to predict categorical data from a given test dataset. First of all, we will implement each of these algorithms separately and then we will use our dataset to train those algorithms. We will then use our test dataset to test these algorithms (KNN, K-mean, Naive bayes SVM and logistic regression). We will be finding the accuracy and efficiency of each algorithm and compare them. Then, we will find the best suited algorithm for this purpose.

ii) Introduction

a.) Motivation

Increasing number of heart diseases have caused a major threat to humans. Due to lack of early diagnosis the patients don't get any treatment which causes serious consequence and sudden deaths. 3 We are planning to use some of these classification algorithms of supervised learning to predict heart disease and find the best algorithm for this purpose. The supervised and unsupervised learning is becoming increasingly popular these days. More and more classification and clustering algorithms are becoming popular these days. We use these algorithms on a given dataset and train our algorithm to recognize new data and put them in best suited class or cluster.

b.) Significance

Most heart attacks remain undiagnosed and many people die during their first heart attack. Many medical instruments are available in the market for the prediction of heart diseases but there are some drawbacks of these instruments like they are very costly, they are not efficient enough for predicting heart diseases. So it is essential to have a frame work that can effectually recognize the prevalence of heart disease in thousands of samples instaneously.

c.)Scope and applications

From the existing system we will be analysing all types of heart disease diagnosis. We have classified several techniques mainly the various classification algorithms. From the analysis we will prove that classification based techniques contribute high effectiveness and obtain high accuracy compare than the previous methods. If these models are implemented, they will provide an end user support and consultant service to patients with easy, quick and accurate diagnosis of heart diseases. Further it will give an engaging self-service experience to the patients.

iii) Literature Survey

Support Vector Machine(SVM) and integer-coded genetic algorithm (GA) were used for heart disease classification by Sumit Bhatia, Praveen Prakash, and G.N. Pillai [1]. Simple Support Vector Machine (SSVM) algorithm was used to determine the support vectors in a fast, iterative manner. An integer-coded genetic algorithm was applied to Cleveland heart disease database for selecting the important and relevant features and discarding the irrelevant and redundant ones.

Prediction of heart disease using neural network was proposed by Dangare et al. in [2]. Feature selection is used to predict the disease. Their method obtained an accuracy of 92.5% for 13 features and 100% accuracy with 15 features. There is a 7.5% improvement after discarding 2 features from 15 to 13.

Jabbar et al. proposed a method using associative classification and feature subset selection for risk score of disease [3]. Authors used information gain, symmetrical uncertainty, and genetic algorithm as feature selection measures. Their method obtained an accuracy of 95% with hybrid feature selection. Heart disease data set collected with 11 features for experimental analysis.

Assessment of coronary heart events risk factors was proposed by karaolis et al. [5]. Authors investigated 2 types of risk factors namely modifiable and non-modifiable. 528 samples were collected and data mining analysis was done using C4.5. The Highest accuracy obtained by their model was 75% for PCI and CABG models. Authors used C4.5 classifier without feature selection measures. The accuracy obtained by this approach is less compared with other approaches.

Diagnosis of heart disease using regression trees was proposed by Amir [6]. Authors collected 116 heart sound signals data set and applied regression tree. Their model is proposed to classify Phonocardiogram's (PCG) data. Authors calculated like hood ratio to classify the disease. Their method obtained an accuracy of 99%. In the year 2014 authors proposed a framework to expect the coronary illness using multilayer perceptron. Their method uses 13 clinical elements as input and achieved an accuracy of 98%. Literature mentioned in this related work has not used effective feature selection measures to improve the accuracy. Authors used weak classifiers to predict the disease. In this paper, we integrated PSO with KNN classifier to obtain effective results.

<u>Sr no.</u>	<u>Authors and Year (Reference)</u>	<u>Title (Study)</u>	<u>Concept / Theoretical model/ Framework</u>	<u>Methodology used/ Implementation</u>	<u>Limitations / Future Research/ Gaps identified</u>
[1]	S.Bhatia, P. Prakash G.N.Pillai 2008	SVM Based Decision Support System for Heart Disease Classification	Classification using simple svm	Svm techniques	Less accuracy in, some cases
[2]	Dangare A. 2012	A data mining approach for predicting heart disease using neural networks	Feature selection	Neural networks	More complex algorithm
[3]	Jabbar MA, Deekshatulu BL, Priti C. 2012	Prediction of risk score for heart disease using associative classification and hybrid feature subset selection	associative classification and feature subset selection	information gain, symmetrical uncertainty	Better approach can be used
[4]	Masethe HD 2014	A Hybrid Data Mining Model to Predict Coronary Artery Disease Cases Using	data mining analysis using c4.5	PCI and CABG models	Methods can be Generalized To be used for any heart disease prediction.

		Non-Invasive Clinical Data			
[5]	Amir M. 2013	Early diagnosis of heart disease using classification and regression trees	Classification and regression trees	Classification And regression trees	Methods can be integrated with other approaches for better results.
[6]	Fahd Saleh Alotaibi	Implementation of Machine Learning Model to Predict Heart Failure	Data preprocessing, model selection, implementation using rapid miner, performance	research used five machine learning models Rapid Miner, is an open source software is used,	proposed and implemented a machine learning model by combining five different algorithms. Results.
[7]	Shrutiko Thadia 2021	Classification algorithms in Python – Heart Attack Prediction and Analysis	Different mathematical techniques are applied	Various implementation techniques are tested and compared	After implementing their accuracy, we can conclude that for this dataset <i>Logistic Regression Classifier</i> is the appropriate model to be used.

[8]	Anjan nikhil repaka, Sai Deepak Ravikanti, Ramya G Franklin	Design And Implementi ng Heart Disease Prediction Using Naives Bayesian	Implementin g Heart Disease Prediction Using Naives Bayesian	The research focuses on heart disease diagnosis by considering previous data and information. To achieve this SHDP (Smart Heart Disease Prediction) is built via Navies Bayesian in order to predict risk factors	The output reveals that the established diagnostic system effectively assists in predicting risk factors concerning heart diseases.
[9]	C sawmiya, P sumitra	Analytical study of heart disease diagnosis using classificatio n techniques	Using Apriori algorithm and SVM (support vector machine) in heart disease prediction.	SVM techniques.	From the analysis it have proved that classification based techniques contribute high effectiveness and obtain high accuracy compare than the previous methods.But this still needs to be improved.

[10]	C. Beulah Christalin Latha	Improving the accuracy of prediction of heart disease risk based on ensemble classification techniques	The machine learns patterns from the existing dataset, and then applies them to an unknown dataset in order to predict the outcome.	ensemble classification techniques	The performance of the process was further enhanced with a feature selection implementation, and the results showed significant improvement in prediction accuracy.
[11]	Garima Choudhary, Shailendra Narayan Singh	Prediction of Heart Disease using Machine Learning Algorithms	Analysis and visual analytics	Decision tree algorithm has been used to make the predictions whether a person has heart sickness or not	It relies on careful evaluation of various medical and pathological facts of the sufferers obtained through health workers, in a complex manner.
[12]	C. Sowmiya, Dr P Sumitra 2017	Analytical Study of Heart Disease Diagnosis Using Classification Techniques	Condition read and check data calculate value time domain frequency domain and non-linear analysis results	Classification, cluster, feature selection , association rule	all types of heart disease diagnosis is analysed classification based techniques contribute high effectiveness and obtain high accuracy compare than the previous methods

[13]	Sakshi Goel, Abhinav Deep, Shilpa Shrivastava, Aparna Tripathi 2019	Comparative Analysis of various Techniques for Heart Disease Prediction	comparing different soft computing techniques to predict Heart Disease based on a combination of risk factors.	Various techniques analysed, literature survey done, different algorithms studied.	machine learning and artificial neural networks provide the most accurate and reliable results..
[14]	Youness Khourdifi, Mohamed Bahaj 2018	Heart Disease Prediction and Classification Using Machine Learning Algorithms Optimized by Particle Swarm Optimization and Ant Colony Optimization	Feature selection optimized techniques such as Particle Swarm Optimization and Ant Colony Optimization	training dataset is prepared From it, features are selected, after that the best subset of features is optimized by combined PSO/ACO algorithm and then, by using the selected best features, new features are classified	Each algorithm worked better in some situations and worse in others. automatic learning and it can be extended for future research.
[15]	Koushik Chandra Howlader, Md. Shahriar Satu, Arup Mazumder 2017	Performance Analysis of Different Classification Algorithms that Predict Heart Disease Severity in Bangladesh	Data preprocessing, feature selection and extraction, classification	collect heart disease records from different hospitals in Bangladesh correlation feature selection (CFS) evaluation method with best first search (BFS) to extract relevant attributes	several classification algorithm are applied, Naive Bayes as better classifier. Future work. In future, will try to get more accurate result with latest heart disease dataset using recent learning approach.

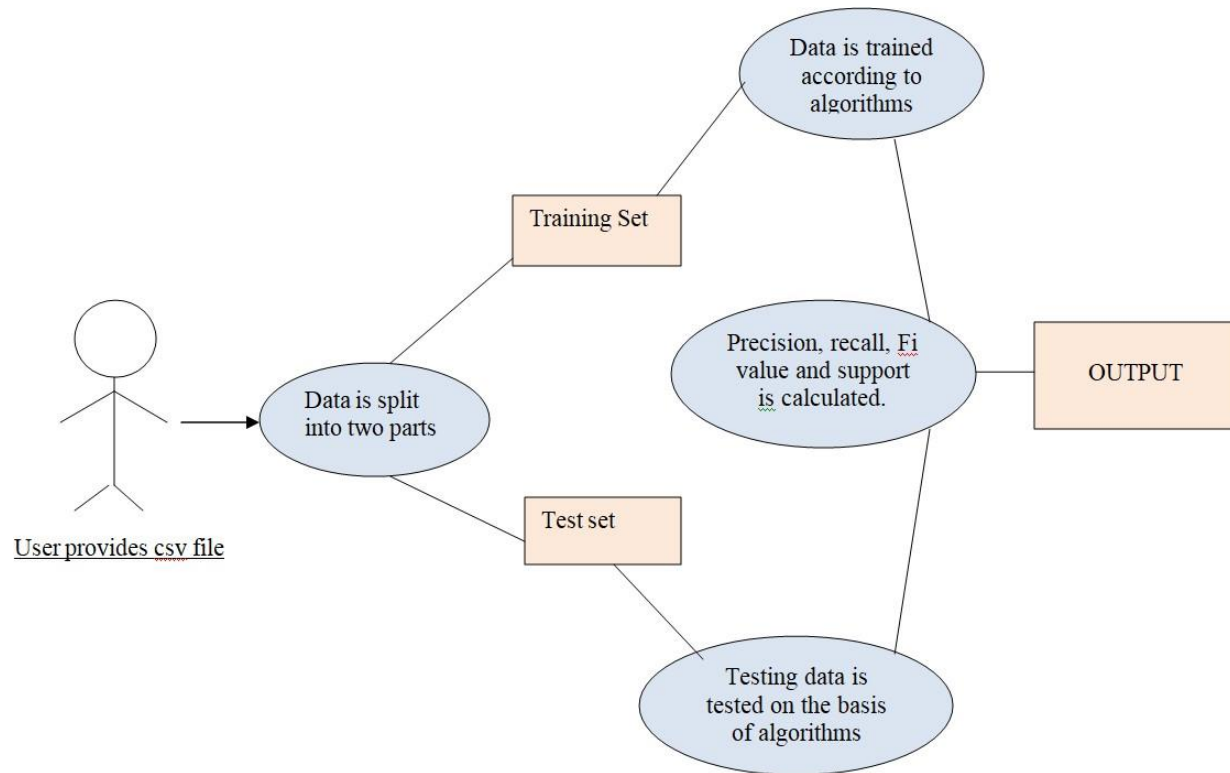
[16]	Divya Annepu Gowtham G 2019	Cardiovascular Disease Prediction Using Machine Learning Techniques	Using a GUI we give in specific details to calculate the level of severity of heart diseases. tried to achieve accuracy with less attributes.	In this study, discussion about the modules of the project which mainly involves training of the dataset and then the model, testing the model and predicting the Output.	work identifies the possibility of heart failure with intelligent ML techniques. accuracy obtained is 97.56%. Interactive platform can be built to communicate about risks and Factors.
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b.) Gaps Identified:

- In cases where the number of features for each data point exceeds the number of training data samples, the **SVM** will underperform.
- With large data, the prediction stage might be slow.
Sensitive to the scale of the data and irrelevant features. (**KNN**)
- This algorithm faces the 'zero-frequency problem' where it assigns zero probability to a categorical variable whose category in the test data set wasn't available in the training dataset. (**Naïve Bayes**)
- It can only be used to predict discrete functions. Hence, the dependent variable of **Logistic Regression** is bound to the discrete number set.

iv)Implementation

a.)Architecture /Framework/ Flowchart:



Methodology Adapted: The main task in this project is to perform supervised learning on the selected data set (in form of csv file). To achieve the goal, we split the data set into two sets namely, test set and the training set and then use the training set to train our algorithms and test set to analyse the results. The main steps in methodology are:

- Selection of proper data set
- Splitting the data set
- Training the algorithms
- Testing of algorithm
- Calculation of accuracy

b.) Algorithms:

I. KNN

```
#!/usr/bin/env python
# coding: utf-8
# In[1]:

import pandas as pd
import numpy as nm
import matplotlib.pyplot as mtp

# In[2]:

df=pd.read_csv('D:\datasets\health\heart.csv')
df.head()

# In[3]:

#extracting dependent and independent variable
x=df.iloc[:,0:13]
print(x.head())
y=df.iloc[:,13]
print(y.head())

# In[4]:

from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test= train_test_split(x, y, test_size= 0.25, random_state=0)
print(x_train.head(),y_train.head())
print(x_test.head(),y_test.head())
# In[5]:

#we need to scaling for accurate predictions ,only for indepedent variables because target is 0 or 1
from sklearn.preprocessing import StandardScaler
st_x= StandardScaler()
x_train= st_x.fit_transform(x_train)
x_test= st_x.transform(x_test)
print(x_train)
print(x_test)

# In[6]:

#fitting KNN classifier to our training data
from sklearn.neighbors import KNeighborsClassifier
classifier= KNeighborsClassifier(n_neighbors=5, metric='minkowski', p=2 ) #we are taking the k-
value as 5 ,p=2 means euclidean distance
classifier.fit(x_train, y_train)
```

```

# In[29]:

#now lets predict the output
y_pred= classifier.predict(x_test)
# In[30]:

print(y_pred)

# In[10]:

print(x_test)

# In[22]:

array=nm.array([x_test[0]])
array

# In[25]:
#y_pred= classifier.predict(array)
# In[27]:
#y_pred[0] #output of the first row in the x_test 2darray says the person has heart attack
# In[31]:

#lets find the confusion matrix
from sklearn.metrics import confusion_matrix
cm= confusion_matrix(y_test, y_pred)
cm

```

II. SVM

```

#!/usr/bin/env python
# coding: utf-8
# In[1]:

import pandas as pd
import numpy as nm
import matplotlib.pyplot as mtp

# In[2]:

df=pd.read_csv('D:\datasets\health\heart.csv')
df.head()

# In[3]:

#extracting dependent and independent variable
x=df.iloc[:,0:13]
print(x.head())

```

```
y=df.iloc[:,13]
print(y.head())
```

```
# In[4]:
```

```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test= train_test_split(x, y, test_size= 0.25, random_state=0)
print(x_train.head(),y_train.head())
print(x_test.head(),y_test.head())
```

```
# In[5]:
```

```
#we need to scaling for accurate predictions ,only for indepedent variables because target is 0 or 1
from sklearn.preprocessing import StandardScaler
st_x= StandardScaler()
x_train= st_x.fit_transform(x_train)
x_test= st_x.transform(x_test)
print(x_train)
print(x_test)
```

```
# In[6]:
```

```
from sklearn.svm import SVC # "Support vector classifier"
classifier = SVC(kernel='linear', random_state=0)
classifier.fit(x_train, y_train)
#In[7]
```

```
#Predicting the test set result
y_pred= classifier.predict(x_test)
y_pred
```

```
# In[8]:
```

```
#Creating the Confusion matrix
from sklearn.metrics import confusion_matrix
cm= confusion_matrix(y_test, y_pred)
cm
```

III. NAÏVE BAYES

```
#!/usr/bin/env python
# coding: utf-8
```

```
# In[1]:
```

```
import pandas as pd
import numpy as nm
import matplotlib.pyplot as mtp
```

In[2]:

```
df=pd.read_csv('D:\datasets\health\heart.csv')  
df.head()
```

In[3]:

```
x=df.iloc[:,0:13]  
print(x.head())  
y=df.iloc[:,13]  
print(y.head())
```

In[4]:

```
from sklearn.model_selection import train_test_split  
x_train, x_test, y_train, y_test= train_test_split(x, y, test_size= 0.25, random_state=0)  
print(x_train.head(),y_train.head())  
  
print(x_test.head(),y_test.head())
```

In[5]:

```
from sklearn.preprocessing import StandardScaler  
st_x= StandardScaler()  
x_train= st_x.fit_transform(x_train)  
x_test= st_x.transform(x_test)  
print(x_train)  
print(x_test)
```

In[6]:

```
from sklearn.naive_bayes import BernoulliNB  
classifier = BernoulliNB()  
classifier.fit(x_train, y_train)
```

In[7]:

```
y_pred = classifier.predict(x_test)
```

In[8]:

```
print(y_pred)
```

In[9]:

```
# Making the Confusion Matrix
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)
cm
```

```
# In[10]:
```

```
from sklearn.naive_bayes import GaussianNB
classifier = GaussianNB()
classifier.fit(x_train, y_train)
```

```
# In[11]:
```

```
y_pred = classifier.predict(x_test)
```

```
# In[12]:
```

```
print(y_pred)
```

```
# In[13]:
```

```
# Making the Confusion Matrix
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)
cm
```

IV. LOGISTIC REGRESSION

```
#!/usr/bin/env python
# coding: utf-8
```

```
# In[2]:
import pandas as pd
```

```
# In[3]:
df=pd.read_csv('D:\datasets\health\heart.csv')
```

```
# In[4]:
df.head()
```

```
# In[5]:
import numpy as nm
import matplotlib.pyplot as mtp
```

```
# In[6]:
#extracting dependent and independent variable
x=df.iloc[:,0:13]
```

```
# In[7]:  
print(x.head())
```

```
# In[8]:  
y=df.iloc[:,13]
```

```
# In[9]:  
print(y.head())
```

```
# In[10]:  
#split the dataset into training and test dataset  
from sklearn.model_selection import train_test_split
```

```
# In[11]:  
x_train, x_test, y_train, y_test= train_test_split(x, y, test_size= 0.25, random_state=0)  
print(x_train.head(),y_train.head())  
print(x_test.head(),y_test.head())
```

```
# In[12]:  
#we need to scaling for accurate predictions ,only for indepedent variables because target is 0 or 1  
from sklearn.preprocessing import StandardScaler  
st_x= StandardScaler()  
x_train= st_x.fit_transform(x_train)  
x_test= st_x.transform(x_test)
```

```
# In[13]:  
print(x_train)  
print(x_test)
```

```
# In[14]:  
#fitting to training set  
from sklearn.linear_model import LogisticRegression  
classifier= LogisticRegression(random_state=0)  
classifier.fit(x_train, y_train)
```

```
# In[15]:  
#Predicting the test set result  
y_pred= classifier.predict(x_test)  
print(y_pred)
```

```
# In[16]:  
#making confusion matrix to test the accuracy  
from sklearn.metrics import confusion_matrix
```

```
# In[17]:  
conf_mat=confusion_matrix(y_test,y_pred)
```

```
# In[18]:  
print(conf_mat)
```

c.) Complexity Analysis:

KNN Algorithm: For the brute-force neighbor search of the kNN algorithm, we have a time complexity of $O(n \times m)$, where n is the number of training examples and m is the number of dimensions in the training set.

training time complexity: $O(1)$

Training space complexity: $O(1)$

Prediction time complexity: $O(k * n * d)$

Prediction space complexity: $O(1)$

SVM Algorithm: Training time for SVM can be given as $O(nd^2)$ if $d < n$

Where d is the dimension

n is the no. of data points

The Run time (predicting time) can be given as $O(kd)$

where k is the no. of support vectors ($0 < k \leq d$)

d is the no of data points.

Naïve Bayes Algorithm: The computational efficiency of Naive Bayes lies in the fact that the runtime complexity of Naive Bayes classifier is $O(nK)$, where n is the number of features and K is the number of label classes.

Logistic Regression: The runtime complexity of Logistic Regression is $O(d)$. Hence, Logistic regression is very good for low latency applications, i.e, for applications where the dimension of the data is small.

the space complexity during runtime is in the order of d , i.e, $O(d)$.

v) Results and Analysis

We trained all the algorithms and tested them against the testing dataset, we calculated the accuracy of each algorithm. To calculate the accuracy, we used various measures like precision, recall and F-measure. Once we had the accuracy for each algorithm, we compared the accuracies and gave the best suited algorithm for the given dataset and further used the algorithm to predict the results for new data. We will be able to tell the best algorithm to predict the classification for the dataset.

Results of our findings are shown in the following tables and graphs-

TARGET	precision	recall	f1-score	support
0	0.73	0.89	0.80	36
1	0.91	0.78	0.84	55
avg/total	0.84	0.82	0.83	91

Table-1 KNN CLASSIFICATION:

Accuracy 0.8241758241758241

Target	precision	recall	f1-score	support
0	0.78	0.65	0.71	49
1	0.66	0.79	0.72	42
avg/total	0.72	0.71	0.71	91

Table2:**Naïvebayes**
Accuracy 0.7142857142857143

Target	precision	recall	f1-score	support
0	0.7	0.86	0.78	36
1	0.89	0.76	0.82	55
weightedavg	0.82	0.8	0.8	91

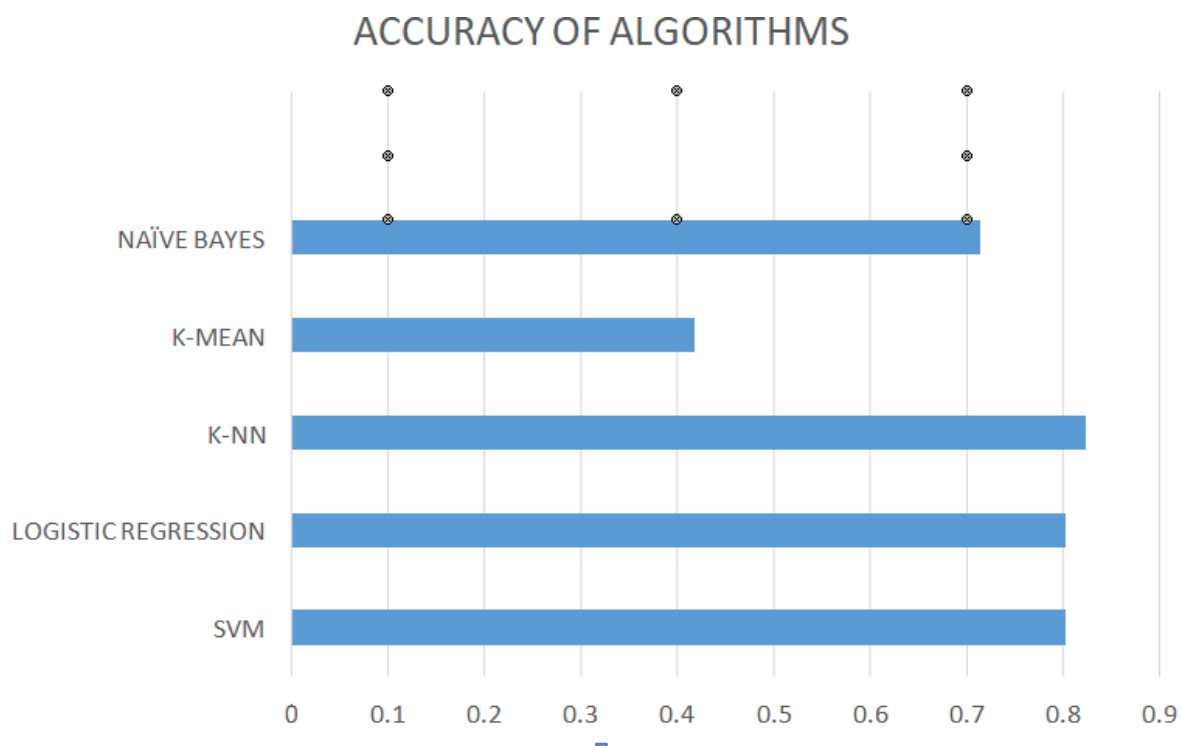
Table3:**SVM**
Accuracy-:0.8021978021978022

Target	precision	recall	f1-score	support
0	0.73	0.84	0.78	38
1	0.87	0.77	0.82	53
weightedavg	0.81	0.8	0.8	91

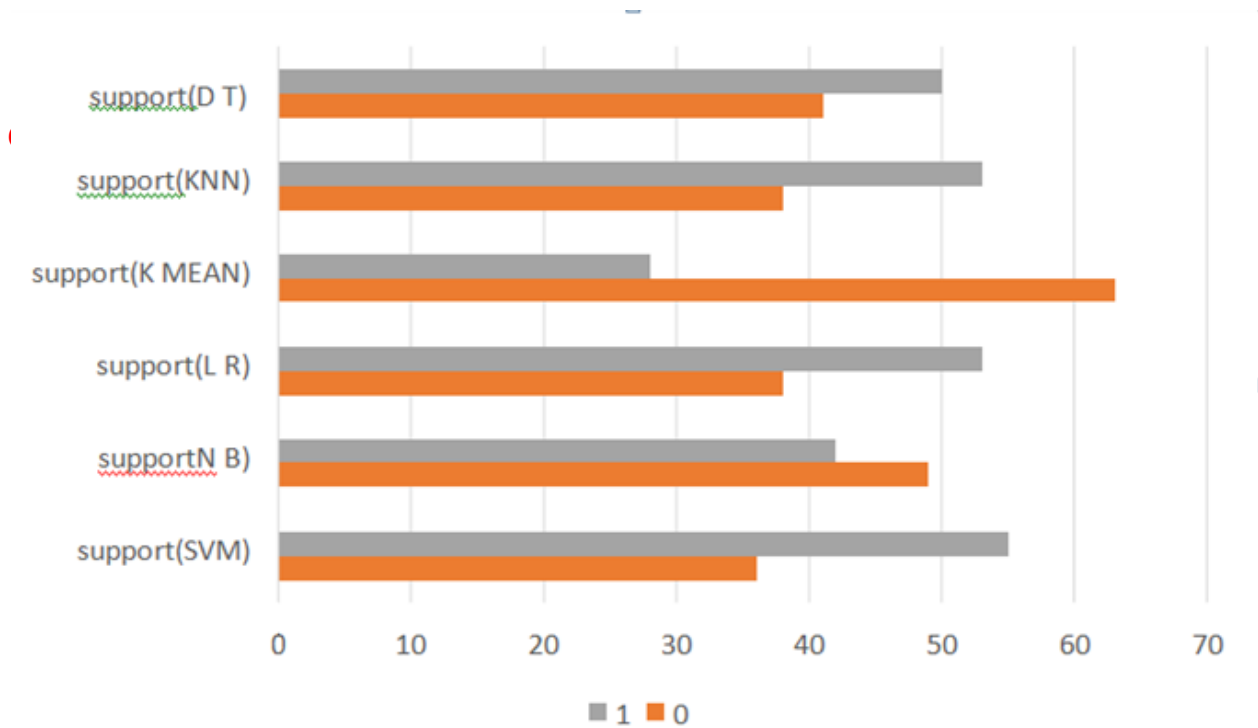
Table4:LogisticRegression

Accuracy-:0.8021978021978022

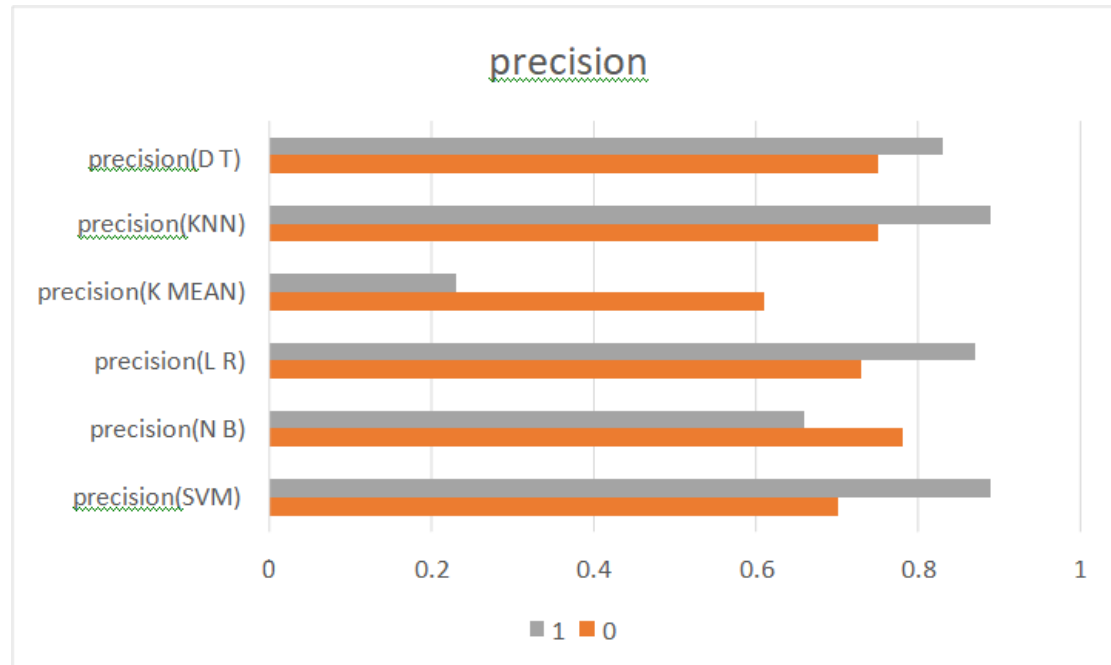
➤ Outcome in-terms of Graphs (result):



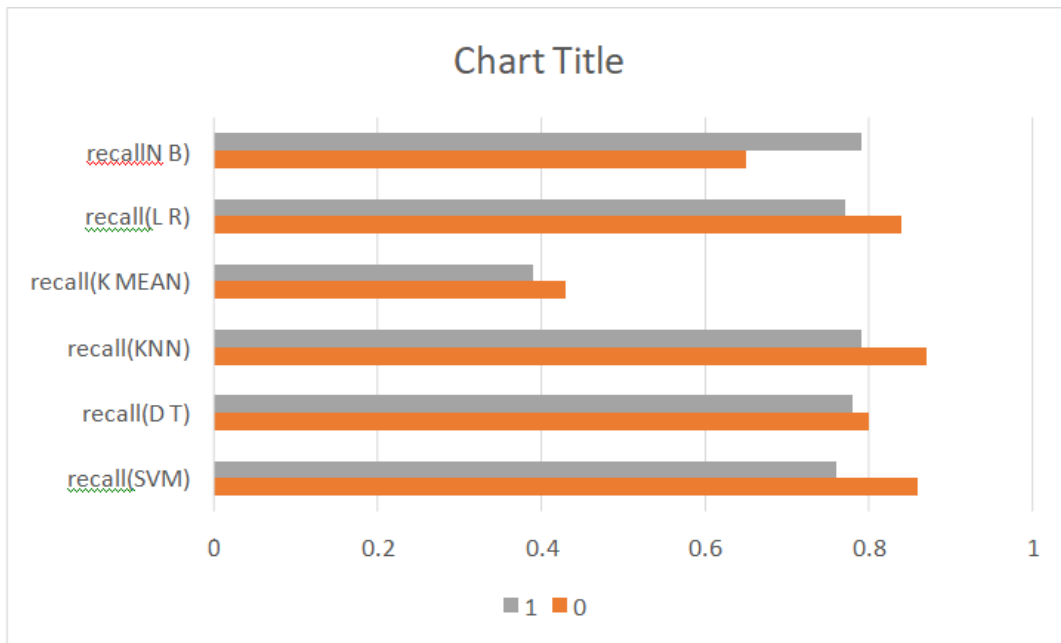
Graph1-:Accuracy comparison of learning



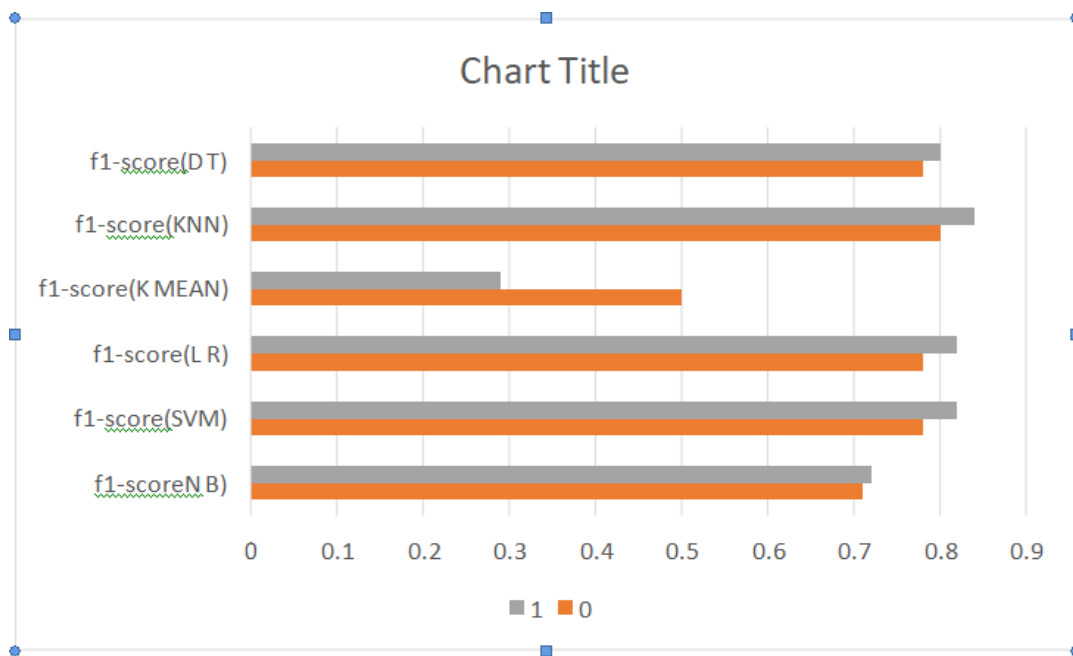
Graph2:-Comparison of support factor



Graph3:-Comparison of precision



Graph4:-Comparison of recall data



Graph5:-f1 score of algorithms

Evaluation metrics

So for evaluating a Classification model, we have the following ways:

1. Log Loss or Cross-Entropy Loss:

- It is used for evaluating the performance of a classifier, whose output is probability value between the 0 and 1.
- For a good binary Classification model, the value of log loss should be near to 0.
- The value of log loss increases if the predicted value deviates from the actual value.
- The lower log loss represents the higher accuracy of the model.

2. Confusion Matrix:

- The confusion matrix provides us a matrix/table as output and describes the performance of the model.
- It is also known as the error matrix.
- The matrix consists of predictions result in a summarized form, which has a total number of correct predictions and incorrect predictions.

The matrix looks like as below table:

n = total predictions	Actual: No	Actual: Yes
Predicted: No	True Negative	False Positive
Predicted: Yes	False Negative	True Positive

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Total Population}}$$

vi.Future Work

The diagnosis of heart disease with automatic learning and it can be extended for future research. There are several limitations to this study, the tools used in this study such as the processing power of the computer and the time limit available for the study. This type of study requires state-of-the-art resources and expertise in the respective fields.

vii) References:

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